Effect of Family Support on Survival Experience among Elders with Alzheimer's Disease and Related Dementias (ADRD)

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Research Questions

Do elders without family live shorter than elders with family among those diagnosed with ADRD?

Introduction

➢ Background

- 5.4 million people with ADRD in 2014
  - Projected to be 15 million by 2060

(Centers for Disease Control and Prevention)

- Little is known about situations of elders with ADRD who do not have family support

➢ Purpose

- Assess survival experience among elders without family and elders with family

➢ Study Design

- Secondary analysis
- Adult Changes in Thought (ACT) study at Kaiser Permanente Washington
- Longitudinal study since 1986

➢ Definitions

- Elders = Adults aged older than 65 years
- Elders without family = Elders who do not have any living spouses or children at the onset of ADRD
- Elders with family = Elders who have living spouses or children at the onset of ADRD

Methods

➢ Study Population

- Participants of ACT study who were diagnosed with ADRD between 2006 and 2016 (N = 848)

➢ Statistical Model

- Cox proportional-hazards model
  - Estimated hazard ratios (HR)
  - Elders without family vs. Elders with family (ref)
  - Adjusted for demographic covariates
  - Demographic covariates

➢ Outcome

- Survival years since the onset

➢ Predictor of Interest

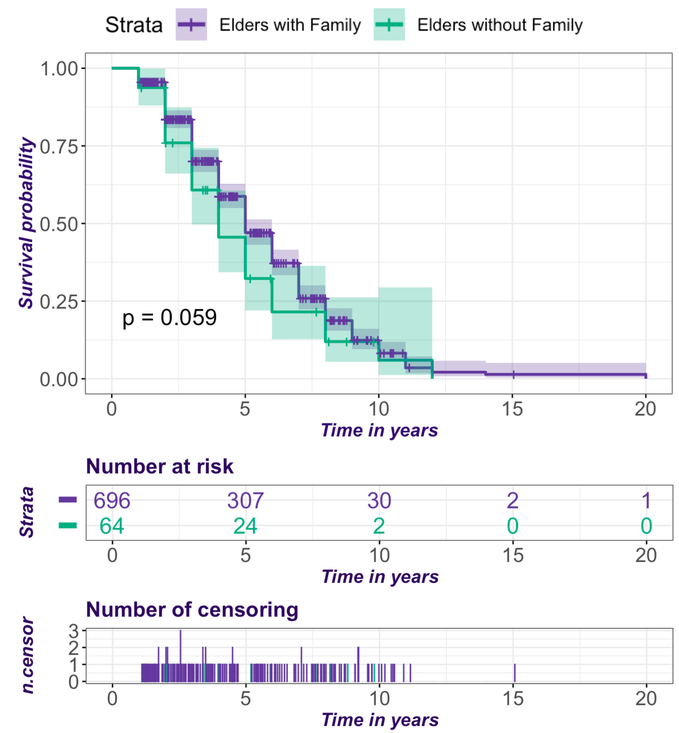
- Family status at the onset of ADRD
  - Having living spouses or children
  - Not having any living spouses or children

➢ Demographic covariates

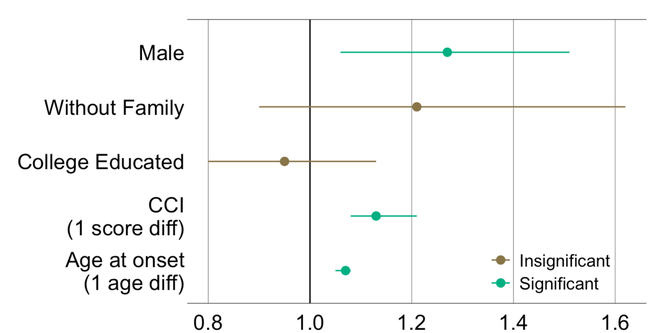
- Age at onset
  - Continuous, one-year difference
- Education level
  - College or more / HS or less (ref)
- Sex
  - Male / Female (ref)
- Charlson Comorbidity Index (CCI)
  - Discrete, one-score difference
- Race/Ethnicity was not included as more than 90% of the participants were non-Hispanic White.

Results

➢ Kaplan-Meier Estimator on Survival Years



➢ Hazard Ratio Estimates by Covariates



Conclusion

Survival experience among elders without family is not statistically different from elders with family. (Cox PH model, *p-value > 0.05*)

- The hazard of death among elders without family is estimated to be 21% higher compared to elders with family.
  (HR = 1.21; 95% CI = [0.90, 1.62])

Discussion

➢ Future research is needed to...

- Assess the social support survey in ACT
  - "How much are you supported by your family, peers, or neighbors?"
- Assess other health outcomes
  - Primary care hospitalization
  - Time to nursing home
- Examine challenges elders without family face in qualitative research

Limitations

➢ Missing values

- 11.6% of the total participants (848)

➢ Limited sample size

- Elders without family n = 64 (8.4%)
- Elders with family n = 696 (91.6%)

➢ Lack of generalizability

- Non-Hispanic White who have a membership at Kaiser Permanente